Low-Cost Geometric Directions in Frozen Speech Models: Where a Training-Free Deepfake-Hardness Signal Helps—and Where It Does Not

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Abstract

Deepfake-speech detectors are inconsistent in two systematic ways: against a fixed detector some synthesis systems evade detection far more often than others, and a detector trained on one recording domain often fails on another. Working entirely at small scale—one GPU, a frozen ∼95M-parameter encoder, no detector retraining—we show that a meaningful, screen-surviving fraction of this inconsistency is legible in the *geometry* of the frozen representation (R^2=0.277, ≈32% of the reliability-corrected ceiling). Inside the encoder there is a single natural-versus-synthetic direction, and a system's position and spread along it track how hard it is to detect: the link recurs within every corpus we test, though *which* statistic carries it differs—spread ($\mathrm{sd}_{\mathrm{along}}$) on the 61-system MLAAD, where it is the strongest of 83 quantities and the only one to survive a global false-discovery-rate correction; position on the few-system ASVspoof/In-the-Wild corpora (a pre-registered cross-corpus transfer was inconclusive, ρ=0.25, p=0.41, n=13). Two results follow. The direction rotates across datasets to near-orthogonality, a geometric correlate of why detectors fail to transfer. And it is a detector-training-free corrective signal conditional on headroom: it cuts the error of a deliberately undertrained detector by ≈21% (and, in-dataset on MLAAD, by ≈40%), and repairs a *naturally* weak cross-domain detector (AASIST) by 26–33 EER points, while leaving an already-strong one unchanged. Every quantitative claim is independently recomputed.

1 Introduction

Synthetic speech is realistic enough that its detection is a security necessity, and the standard defence flags synthetic utterances with a classifier. Yet detectors are unreliable in two ways aggregate accuracy hides: some synthesis systems evade a fixed detector far more often than others (Wang et al., 2020; Müller et al., 2024a), and a detector tuned on one recording domain often fails on another (Müller et al., 2022). Neither had a clean explanation.

We provide one, at small scale. Modern detectors sit on top of a large frozen self-supervised (SSL) speech encoder (Chen et al., 2022; Tak et al., 2022; Wang & Yamagishi, 2022); we study that frozen representation before any decision boundary is drawn. This follows the linear-representation-hypothesis program in language models, where concepts like gender bias or refusal are recoverable as single frozen-representation directions (Bolukbasi et al., 2016; Arditi et al., 2024); we ask whether an analogous concept direction organizes a security-relevant failure mode in speech encoders. Our central finding is a **natural-versus-synthetic axis**: a system whose utterances stay tight along it is detected easily, one whose utterances spread along it is hard (Fig. 1); that spread, $\mathrm{sd}_{\mathrm{along}}$, is our predictor. Three results follow. (i) *The association:* $\mathrm{sd}_{\mathrm{along}}$ predicts per-system hardness, the strongest of 83 quantities and the only survivor of global multiple-testing correction (Sec. 3.2). (ii) *The axis rotates:* each dataset has its own axis, so a transferred one is uninformative—a geometric correlate of why transfer fails. (iii) *A detector-training-free corrective signal:* fusing the axis into a score

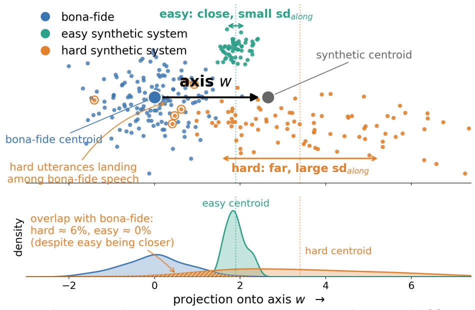


Figure 1: **Schematic (simulated 2-D data).** *Top:* utterance embeddings and the axis w. An easy system stays tight along w; a hard one spreads across the boundary, so some utterances land in the bona-fide region and evade (circled). *Bottom:* projections onto w—the hard system leaks $\approx 6\%$ of its mass into bona-fide despite sitting *farther* out. Spread (sd$_{\text{along}}$), not mean position, predicts hardness.

needs no detector training and no difficulty labels, and helps only under headroom (Sec. 4). Every number is independently recomputed (Sec. 3.3).

2 Setup

Detector. Our primary detector (WavLM-GAT) is the phoneme-level graph-attention model of Zhang et al. (2025): a frozen WavLM encoder feeds adaptive phoneme pooling, a graph-attention network (Veličković et al., 2018), a BiLSTM, and a linear head. The encoder is frozen ($\sim$95M parameters; WavLM-base, 94.7M); only the $\sim$13M head trains. For architecture generality we also use AASIST (Jung et al., 2022), a spectro-temporal detector *not* built on WavLM.

Corpora and hardness. The association is established on MLAAD (Müller et al., 2024b) (61 TTS systems, ≥ 8 utterances each); the lever and rotation checks add ASVspoof 2019 LA (Wang et al., 2020), ASVspoof 2021 (Yamagishi et al., 2021), and In-the-Wild (Müller et al., 2022). Per-system hardness is $1 - \text{AUC}$ of the detector's scores for that system against a shared bona-fide pool. Class overlap is exactly what lowers AUC, so a geometric "spread across the boundary" should map onto hardness.

Objective hardness. Hardness is a stable target: three seeded WavLM-GAT detectors agree on the per-system ranking at $\rho = 0.89$–0.97 on MLAAD (App. A.3). **Compute.** Everything runs on one GPU; the largest run is $\approx 10^{16}$ FLOPs (under the 10^{20} budget and 3B-parameter cap).

3 Experiments

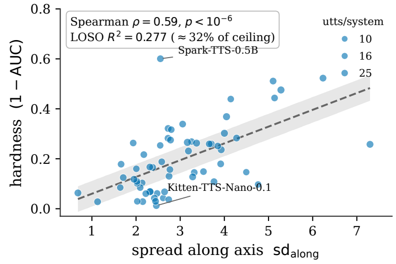
3.1 The axis and its spread

Frozen embeddings. For each utterance (16 kHz, 3 s) we mean-pool the frozen WavLM layer-12 hidden states over time, one 768-d vector per clip. Among the layers we probed (0, 9, 12) the deepest carries the strongest geometry signal (App. A.1), consistent with SSL encoders stratifying acoustic detail from linguistic content with depth (Pasad et al., 2021; 2024). Each coordinate is standardized by the training bona-fide mean and SD.

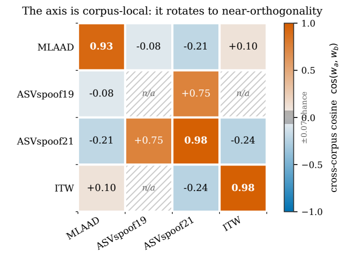
The axis. Per corpus we estimate the direction as the standardized centroid difference $w = \text{centroid}(\text{synthetic}) - \text{centroid}(\text{bona})$—a difference-of-means concept direction (Bolukbasi et al., 2016; Arditi et al., 2024), not a trained classifier: it uses the bona/synthetic *group* labels but no fitted boundary and no per-system difficulty labels. We estimate w corpus-internally (directions differ across corpora, Sec. 3.2) and **leave-one-system-out** (LOSO): when scoring a system we rebuild w from the bona pool plus every *other* system's synthetic clips, so a held-out system never informs the axis judging it. For each system we record sd$_{\text{along}}$, the standard deviation of its projections onto w; the mechanism (Fig. 1) predicts this second moment beats mean position.

3.2 Results: the hardness association and its rotation

A natural–synthetic axis predicts detection hardness. Across 61 MLAAD systems, hardness rises with sd$_{\text{along}}$ (Fig. 2a; Spearman $\rho = 0.59$, $p < 10^{-6}$; LOSO $R^2 = 0.277$, permutation



(a) The hardness association.



(b) The axis is corpus-local.

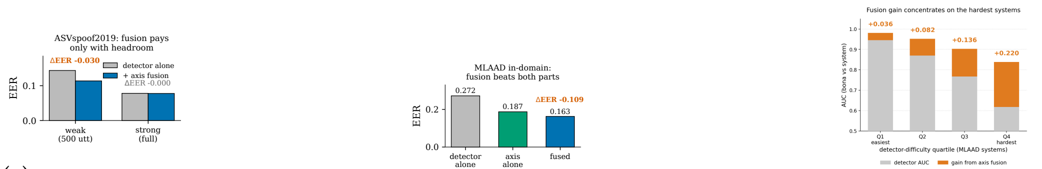
Figure 2: **(a)** Spread along the corpus-internal axis predicts MLAAD hardness. **(b)** Cosine between per-corpus axes (diagonal: within-corpus split-half reliability 0.93–0.98; hatched cell lacks an estimate). Cross-corpus cosines sit near zero or negative, inside/at the ± 0.07 chance band; the lone large positive (ASVspoof-2019/2021, shared source) is the exception that proves the rule.

$p = 2.5 \times 10^{-4}$; bootstrap CI $[0.41, 0.73]$). Systems that straddle the boundary produce utterances that look genuine to the encoder, and those evade. Spread beats every alternative head-to-head. Isotropic (axis-free) total variance—the "diverse systems are just hard" null—predicts poorly (radius of gyration $\rho = -0.31/-0.38$ at layers 12/0); mean position is likewise weak (s_{along} $\rho = -0.26$; centroid norm $+0.29$); the best runner-up is a trajectory-dynamics feature ($\rho = +0.35$). Against sd_{along}'s $\rho = +0.60$, none survives correction: of the **83** quantities tested, sd_{along} is the *only* global Benjamini–Hochberg survivor (Benjamini & Hochberg, 1995) ($q \approx 3 \times 10^{-5}$; App. A.1). It strengthens as measurement noise falls ($\rho = 0.59/0.64/0.69$ at $\geq 8/12/16$ utterances/system)—the attenuation signature of a real effect—and is robust to jackknifing and loudness control (full robustness statistics in App. A.4).

The link replicates within every corpus—through different axis statistics. The axis-geometry$\to$hardness link recurs within each corpus, but which statistic carries it tracks corpus structure. On the 61-system MLAAD it is spread; on the few-system ASVspoof/ITW corpora—where per-attack *position* varies over a 3.5× wider range—it is position: ASVspoof-2019 LA s_{along} $\rho = -0.71$ ($p = 0.006$, 13 attacks); ITW speaker s_{orth} $\rho = +0.53$ ($p = 0.003$, 29 speakers); ASVspoof-2021 LA corpus-internal position $\rho = +0.60/+0.64$ (WavLM-GAT/AASIST), significant only uncorrected (Sec. 3.3). The spread statistic itself is confirmed only on MLAAD; this corpus-dependence of the operative statistic is the axis-rotation finding seen from the hardness side.

Directional, not circular. A logistic/LDA probe fit on the same frozen embeddings—our proxy for the detector's own decision direction—is a stronger *classifier* yet aligns only moderately with our one-parameter axis ($\cos \approx 0.4$ against this detector-proxy probe; App. A.5): the axis is not the detector's logit direction relabeled (Zhang et al., 2021). It also replicates across detector families on AASIST (a near-independent target, ≈ 0.55 correlation with WavLM-GAT's hardness): WavLM sd_{along} *ranks* AASIST hardness ($\rho = +0.35$, $p = 0.006$), though the predictive out-of-sample form does not (LOSO $R^2 = -0.032$, $p = 0.13$).

The axis rotates across datasets. A predictor fit on one corpus fails on another because its direction is not the same elsewhere. Each corpus's internal axis is estimated almost noiselessly (split-half 0.93–0.98, Fig. 2b), yet cross-corpus axes are near-orthogonal: MLAAD–ASVspoof-2019 cosine ≈ -0.08, close to the chance level expected between two random directions in 768 dimensions ($\mathbb{E}|\cos| \approx 0.03$, 95th pct ≈ 0.07; Vershynin, 2018). At this magnitude the pair is best read as *unrelated*, not reliably reversed; the sign turns dependable only for larger-magnitude pairs (MLAAD–ASVspoof-2021 ≈ -0.21, ITW–ASVspoof-2021 ≈ -0.24). Either way a transferred axis carries no usable information and must be re-estimated on the target domain (part of the rotation may be recording-channel nuisance (Chen et al., 2021)).



(a) ASVspoof2019. (b) MLAAD. (c) Gain vs. difficulty.

Figure 3: **(a)** The recipe cuts a weak ASVspoof2019 detector's EER (0.144 → 0.114) but leaves a strong one unchanged. **(b)** In-domain on MLAAD, fusion (0.163) beats detector-alone (0.272) and axis-alone (0.187). **(c)** Per-quartile stacked bars: gray is detector-alone AUC, orange the fusion gain on top (bar height = fused AUC); the gain rises with difficulty, +0.036 (easiest, 0.950 → 0.985) to +0.220 (hardest, 0.617 → 0.837).

3.3 Verification

A red-team pass recomputed every number from cached artifacts, correcting or removing 13 claims (App. A.2 itemizes them). The screen is a global Benjamini–Hochberg procedure over all 83 recorded per-system tests (Benjamini & Hochberg, 1995), permutation p-values throughout (Ojala & Garriga, 2010); $\mathrm{sd}_{\mathrm{along}} \rightarrow$hardness is the sole survivor (the AASIST and channel checks are confirmatory follow-ups, not family members). It is *not* a loudness/channel artifact (partial $\rho = +0.58$ given RMS energy, $p = 8 \times 10^{-7}$; likewise given utterance count), and the target is reliable: the noise-corrected R^2 ceiling is 0.86, so $R^2 = 0.277$ captures ≈ 32% of the *explainable* variance (App. A.3).

4 Axis fusion improves performance under detector headroom

Because the axis identifies *which* systems a detector will struggle with, its projection fuses into the score as $s_{\mathrm{fused}} = z(\mathrm{logit}) + \lambda\, z(\mathrm{axis})$, needing no detector training and no difficulty labels. To avoid leakage, the z-statistics and λ are fit on each system-disjoint fold's *training* partition only.

Headroom decides. A *deliberately undertrained* WavLM-GAT (500 utterances) drops from EER 0.144 to 0.114 under the axis (ΔEER $= -0.030$, 95% CI $[-0.046, -0.013]$; paired bootstrap $p < 10^{-3}$, App. A.5), improving all five folds—a ≈21% reduction (Fig. 3a). The identical procedure on a strong, near-ceiling WavLM-GAT (EER 0.078) does nothing beyond noise (ΔEER $= -0.0004$ to $+0.004$, $p \approx 0.56$–0.80). In-domain on MLAAD, far from ceiling, the same fusion improves EER 0.272 → 0.163 (axis alone 0.187), negative across all 15 fold×seed cells (Fig. 3b).

A naturally weak detector, too. The effect is not an artifact of the crippled detector. AASIST evaluated cross-domain is naturally weak (MLAAD EER 0.376; ITW 0.486), and axis fusion repairs it substantially—MLAAD 0.376 → 0.116 (−26 pts), ITW 0.486 → 0.161 (−33 pts)—while adding nothing where AASIST is strong (ASVspoof-2021 0.073 → 0.076). Honest caveat: this cross-domain axis is a *supervised* in-corpus LDA probe on the frozen embeddings, which *alone* matches or beats the fused score (MLAAD 0.100, ITW 0.101)—so it is in-domain axis information, not a zero-cost free lunch (App. A.5).

Why the axis works. The gain lands on the hardest systems: by MLAAD detector-difficulty quartile, fusion's AUC gain grows monotonically from +0.036 (easiest) to +0.220 (hardest) (Fig. 3c). Hard systems sit close to bona-fide speech yet spread along a direction the detector under-weights; re-injecting it recovers exactly them—a strong detector already encodes it.

5 Limitations and conclusion

TTS-specific, single backbone. On voice conversion all predictors give $\rho \approx 0$ (converted speech inherits content and spread from its source); geometry uses one backbone (WavLM; wav2vec 2.0 (Baevski et al., 2020) and HuBERT (Hsu et al., 2021) untested), with the main association confirmed across three seeds. **Labels required.** The axis is class-labeled and rotates, so a new domain needs its own labels. **Cross-dataset transfer is unconfirmed.** The pre-registered primary test on ASVspoof-2021 LA failed ($\rho = +0.25$, $p = 0.41$, $n = 13$); the exploratory $\rho = +0.60$ variant does not survive family correction (Sec. 3.3). Within these bounds a low-cost frozen direction helps understand and partially correct weak detectors within a dataset. **Reproducibility:** every number regenerates from committed single-GPU scripts over cached embeddings, released with logs and audit.

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A Appendix

A.1 The 83-quantity screen (top competitors)

Every recorded predictor→hardness test across the study (83 tests, spanning the geometry battery, position/spread axis statistics, and trajectory dynamics at layers 0, 9, 12) was enumerated and corrected with a single global Benjamini–Hochberg FDR procedure. Table 1 lists sd_{along} against the reviewer-relevant competitors: isotropic total variance (radius of gyration, the axis-free null), mean position, and the best non-winning geometry feature. All correlations are Spearman on MLAAD (61 systems); q is the global BH-corrected value. Only sd_{along} survives; the full 30-feature geometry battery alone yields *zero* FDR survivors (best $q = 0.078$).

A.2 Verification / audit details

The red-team pass recomputed every number from raw cached artifacts (not the original analysis code), tallying 8 verified, 6 mislabeled, 5 misleading, and 2 contradicted claims. Representative corrections: (i) a cross-corpus cosine reported as $+0.36$ was in fact negative (-0.14 to -0.21); the $+0.362$ was a within-MLAAD $\cos(w_{\text{mean}}, w_{\text{LDA}})$. (ii) An R^2 of 0.316 attributed to sd_{along} was a three-feature model (sd_{along} alone is 0.277). (iii) The two

Quantity	Role	ρ	p	$q_{\text{BH,global}}$
sd_{along}	**spread along axis (winner)**	+0.597	3.7×10^{-7}	3.1×10^{-5}
`vel_entropy` (L9)	trajectory dynamics (best runner-up)	+0.352	0.005	0.072
`top1_frac` (L9)	spectral concentration	+0.332	0.009	0.074
`part_ratio` (L9)	participation ratio	−0.326	0.010	0.077
`vel_entropy` (L12)	trajectory dynamics	+0.310	0.015	0.094
`mu_norm` (L12)	centroid norm (mean position)	+0.289	0.024	0.109
s_{along}	mean projection (mean position)	−0.262	0.042	0.148
`rog` (L0)	radius of gyration (isotropic var.)	−0.381	0.002	0.059
`rog` (L12)	radius of gyration (isotropic var.)	−0.308	0.016	0.094
`rog` (L9)	radius of gyration (isotropic var.)	−0.206	0.112	> 0.1

Table 1: Top of the 83-quantity screen. sd_{along} is the sole global BH-FDR survivor; isotropic variance and mean position both lose decisively.

contradicted claims: a residual causal-compactness effect flipped sign across seeds (seed-level $p = 0.21$) and was dropped, and the ASVspoof-2021 corpus-internal position result, significant uncorrected, does not survive family correction (permutation p 0.034/0.020 → across-family 0.235/0.164). Reported counts elsewhere were also corrected: the ITW speaker analysis is $n = 29$ speakers (not 58).

A.3 Reliability details

Hardness target. Three independently seeded WavLM-GAT detectors agree on the per-system hardness ranking at Spearman $\rho = 0.89$–0.97 on MLAAD (main↔s42 0.968, main↔s1024 0.913, s42↔s1024 0.893). R^2 **ceiling.** Hardness's split-half correlation is 0.759; Spearman–Brown correction (Spearman, 1904) to full length gives 0.863, so the reliability-imposed R^2 ceiling is 0.863 and $R^2 = 0.277$ recovers $\approx 32\%$ of *explainable* variance. **Axis reliability.** The corpus-internal axis is estimated almost noiselessly: split-half cosine 0.926 (MLAAD), 0.982 (ITW), 0.981 (ASVspoof-2021), versus the chance level expected between two random directions in 768 dimensions ($\mathbb{E}|\cos| \approx 0.03$, 95th pct ≈ 0.07).

A.4 Robustness of the MLAAD association

The MLAAD sd_{along} →hardness link ($\rho = +0.60$) survives leave-one-system jackknifing (ρ range 0.58–0.63; 0/61 folds lose significance) and is undiminished after partialling out loudness (partial $\rho|\text{RMS} = +0.58$, $p = 8 \times 10^{-7}$), controls for n-utterances ($\rho|n = +0.58$), and estimator choice (ridge $R^2 \in [0.275, 0.281]$). Dropping the two most influential systems raises it to $\rho = 0.66$. The attenuation sweep is $\rho = 0.59$ (n=61, ≥ 8 utts) / 0.64 (n=51, ≥ 12) / 0.69 (n=28, ≥ 16). A winner's-curse split-half simulation confirms sd_{along} as the discovery-half winner in 76.7% of splits and significant on the confirmation half in 99.2%.

A.5 Frequently anticipated questions

Q1 (isotropic variance). Does raw axis-free within-system variance predict hardness as well? No: radius of gyration is $\rho = -0.31$ (L12) / -0.38 (L0) and fails FDR ($q \geq 0.07$), versus sd_{along} $\rho = +0.60$, $q = 3 \times 10^{-5}$. The directional statistic is not a relabeling of "diverse systems are hard."

Q2 (naturally weak detector). Does fusion help a detector that is weak for real reasons? Yes: AASIST evaluated cross-domain (MLAAD EER 0.376, ITW 0.486) is repaired to 0.116 and 0.161. The caveat is that the operative cross-domain axis is a supervised in-corpus LDA probe on the frozen embeddings, which alone reaches 0.100 (MLAAD) / 0.101 (ITW)—so this is in-domain axis information, not a zero-cost gain.

Q6 (axis vs. trained probe). A fitted LDA/logistic probe on the same embeddings—used here as a proxy for the detector's own decision direction, since WavLM-GAT's logit direction is not directly extractable at the embedding level—is a stronger *classifier* (EER 0.10/0.11 vs. the axis's 0.19) and fuses slightly better (ΔEER −0.16 vs. −0.11), yet aligns only moderately with the one-parameter axis (cos ≈ 0.4 against this detector-proxy probe)—so the axis is not the detector's logit direction relabeled, just the simplest signal that still predicts hardness and corrects a weak detector.

Q7 (layer choice). Layer 12 was not chosen from a dense pre-registered 0–12 sweep; we probed layers 0, 9, 12 and $L12$ carried the strongest axis geometry. We flag this as a limited, not exhaustive, layer comparison.

Q8 (AASIST ρ vs R^2). $\mathrm{sd}_{\mathrm{along}}$ ranks AASIST hardness ($\rho = +0.35$, $p = 0.006$) but the out-of-sample LOSO-R^2 form is null ($R^2 = -0.032$, $p = 0.13$). Both are real: the monotone rank signal replicates across detector families, but does not translate into predictive out-of-sample variance under AASIST. We claim rank replication, not that the predictive law transfers.

Q9 (λ sensitivity). In the headline fusion, λ selected per fold on training partitions takes only two values, $\{1.5, 2.0\}$, across all 5 folds and 3 seeds—low sensitivity. Fusion p-values are paired bootstrap (B=2000); with 0 exceedances we report the resolution floor $p < 10^{-3}$. An independent calibration-size sweep (J3) shows help scaling with how much the detector under-reads the domain axis: a 250/class supervised head cuts MLAAD EER by $\Delta = -0.136$ ($\approx$50%) and ITW by $\Delta = -0.171$ ($\approx$47%), but ASVspoof (already strong) not at all.

Q10 (contradicted claims). The two claims the red-team removed were the residual causal-compactness effect (sign-unstable across seeds, $p = 0.21$) and the "pre-registered" status of the ASVspoof-2021 position success (it does not survive family correction); both are now reported as null/exploratory in the main text.